

Physics of Cognition: Homework 4

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1.1: Fixed Point Attractors

1

$$\begin{aligned}\frac{dx}{dt} &= -\frac{dV}{dx} \\ &= -\frac{d}{dx}\left(\frac{1}{2}x^2\right) = 0 \\ -x &= 0 \\ x &= 0\end{aligned}$$

To check if this point is stable, we check the following:

$$\frac{d^2V}{dx^2} = 1$$

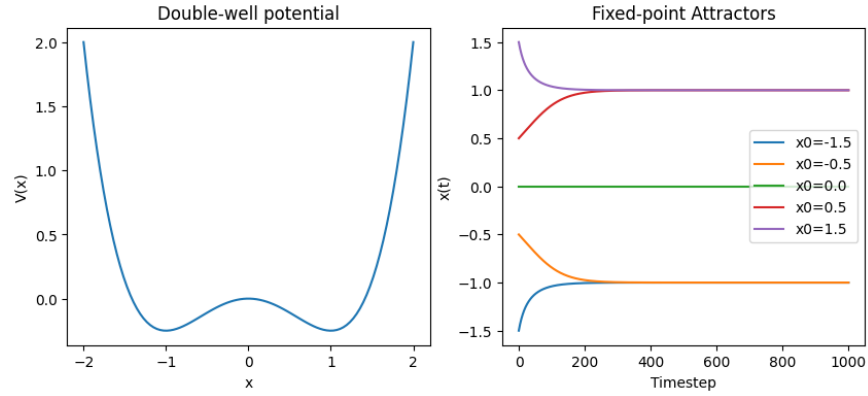
Since this value is positive, the function is concave up at this point and the fixed point is stable.

2

$$\begin{aligned}V(x) &= \frac{1}{4}x^4 - \frac{1}{2}x^2 \\ \frac{dV}{dx} &= x^3 - x = 0 \\ x(x^2 - 1) &= 0 \quad \Rightarrow \quad x = 0, \pm 1\end{aligned}$$

Let's check each point x^* :

$$\begin{aligned}V''(x) &= 3x^2 - 1 \\ V''(0) &= -1 \quad (\text{Unstable}) \\ V''(1) &= 2 \quad (\text{Stable}) \\ V''(-1) &= 2 \quad (\text{Stable})\end{aligned}$$



- $x^* = 0$: $x_0 = 0$
- $x^* = -1$: $x_0 = -1.5, -0.5$
- $x^* = 1$: $x_0 = 0.5, 1.5$

For the unstable fixed point $x^* = 0$, the decision boundary is at $x_{\text{boundary}} = 0$.

3

The stable fixed points serve as memories: depending on the proximity of initial states to these points, trajectories converge cleanly. However, the x -coordinate of the unstable state serves as the threshold determining which stable state the trajectory converges to (i.e., the decision boundary).

1.2: Limit Cycle Attractors

1

$$\begin{aligned} \frac{dx}{dt} &= y = 0 \\ \frac{dy}{dt} &= \mu(1 - x^2)y - x = 0 \\ x &= 0 \\ (x^*, y^*) &= (0, 0) \end{aligned}$$

We then find the Jacobian:

$$\begin{aligned} J &= \begin{pmatrix} 0 & 1 \\ -2\mu xy - 1 & \mu(1 - x^2) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix} \end{aligned}$$

Now we'll calculate the eigenvalues.

$$J = \begin{pmatrix} -\lambda & 1 \\ -1 & \mu - \lambda \end{pmatrix}$$

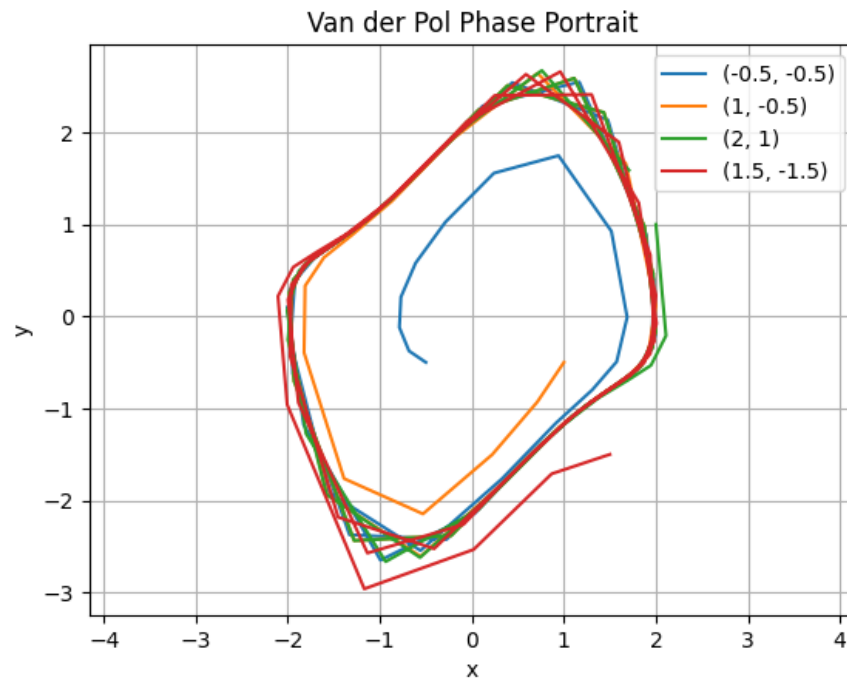
$$0 = -\lambda(\mu - \lambda) + 1$$

$$= -\mu\lambda + \lambda^2 + 1$$

$$\lambda = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}$$

$\lambda < 0$ only if $\mu < 0$. Since the nonlinearity parameter is always greater than 0, the origin is always unstable.

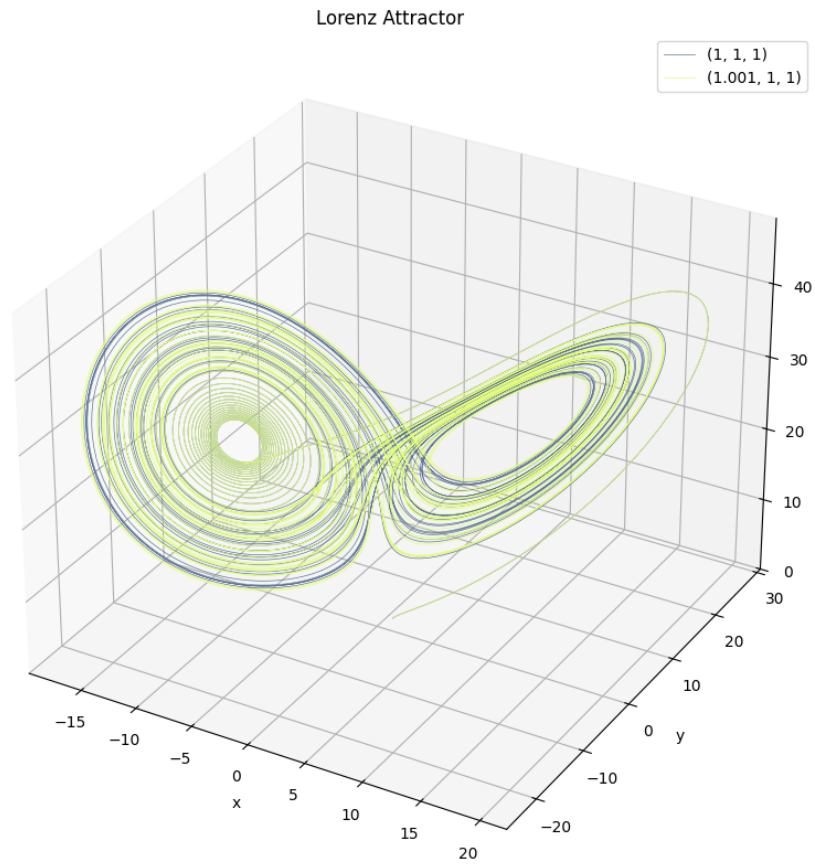
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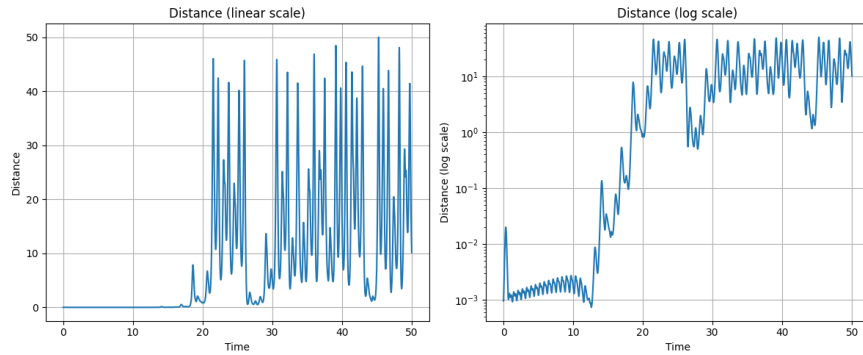
1.3: Chaos

1

Amazing plot.



We can also see the distance increasing exponentially up to a point.



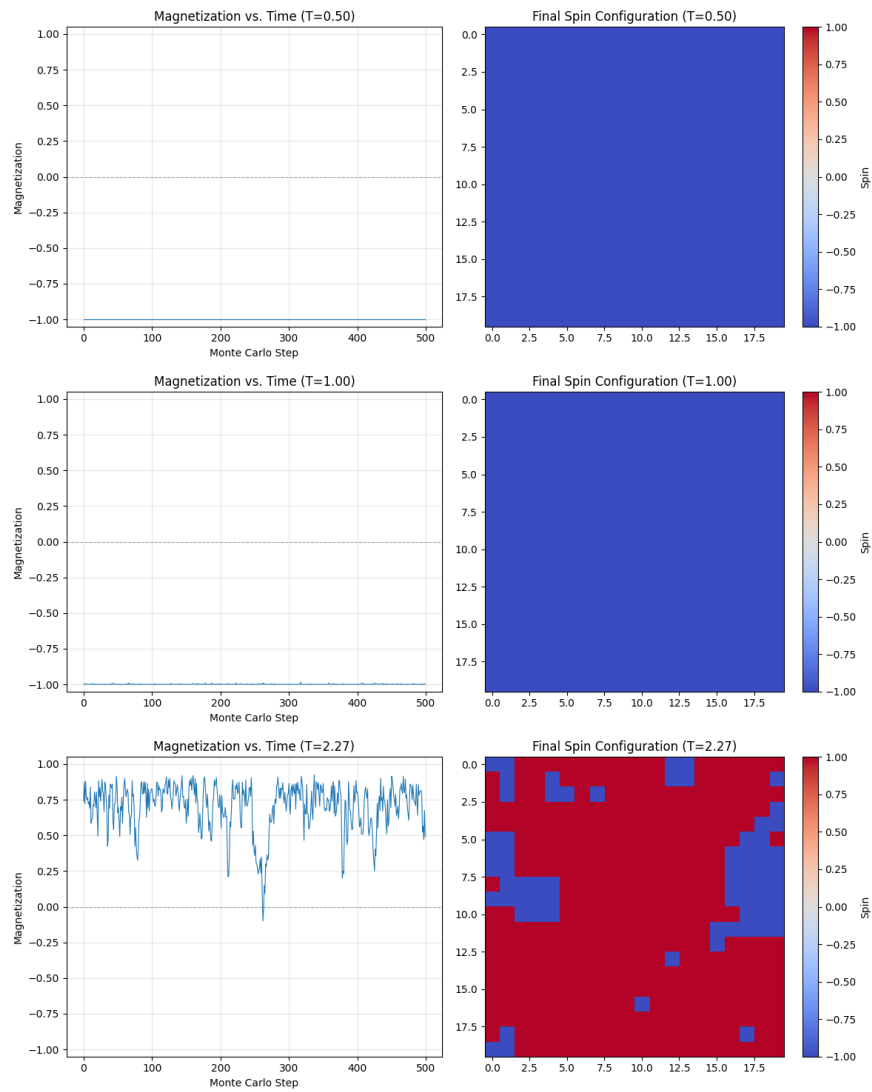
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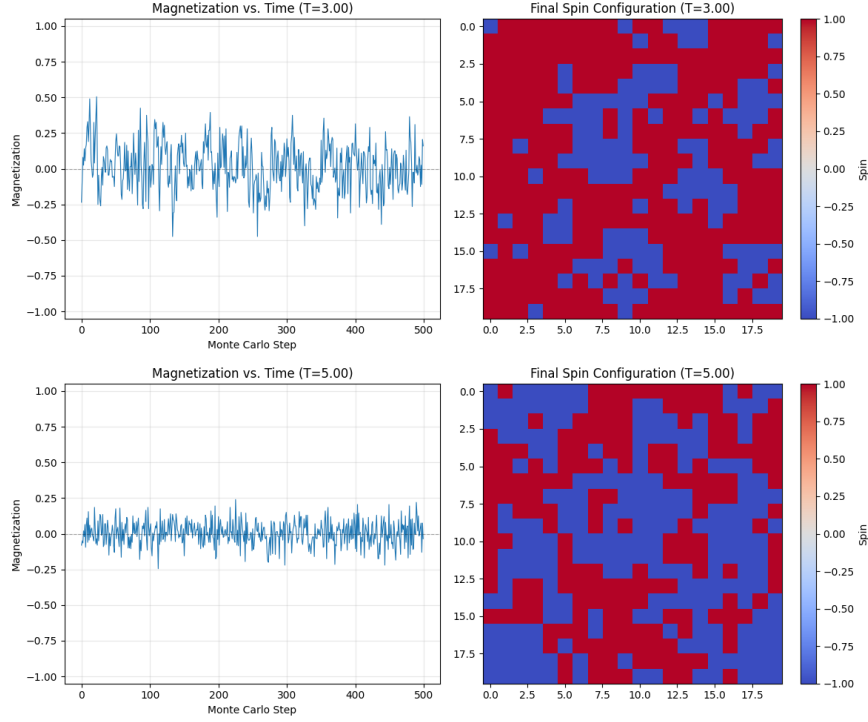
During neural avalanches, the brain exhibits "criticality" where every neuron activation causes exactly one other to activate, resulting in a propagation that

can slow down or continue running. This is very similar to chaotic systems where a minor decrease in criticality quickly kills all the activations but the opposite change can cause an exponential explosion in the brain. This strongly relates to the physics models we've been using where some critical point determines the chaos of the system.

2.1: Ising Model

The configurations with all the spins up or down have the lowest energy since all the "magnets" align perfectly.





$T = 2.27$ is the critical temperature for the 2D Ising model, marking a phase transition between ordered and disordered states. Below T_c , thermal fluctuations are too weak to overcome the energy cost of flipping aligned spins, causing the system to settle uniformly. Above T_c , thermal energy dominates over spin interactions, and the system behaves randomly with uncorrelated spins. At the critical temperature, the system exhibits scale invariance: the correlation length $\xi \rightarrow \infty$, meaning spin correlations extend across the entire lattice without any characteristic length scale.

2.2: Hopfield Networks

1

$$h_1 = J_{12}s_2 + J_{13}s_3 = 2/3$$

$$h_2 = J_{21}s_1 + J_{23}s_3 = -2/3$$

$$h_3 = J_{31}s_1 + J_{32}s_2 = 2/3$$

The signs match the signs of the memory $\xi = (+1, -1, +1)$, so these are stable states.

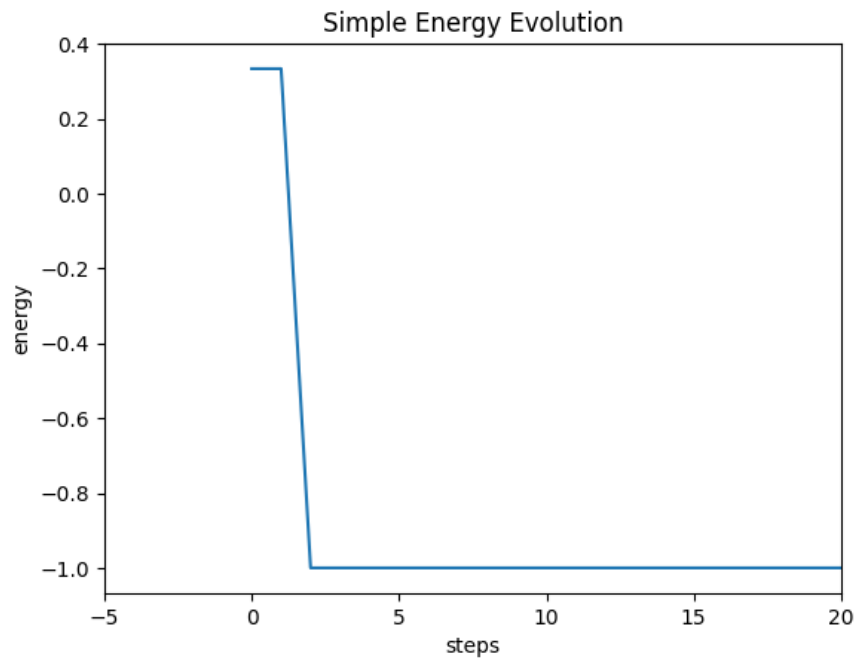
If neuron 2 becomes +1, we error-correct:

$$\begin{aligned}\text{sign}(h_2) &= \text{sign}(-2/3) = - \\ - &\rightarrow +1 = -1\end{aligned}$$

Yay! $\{+1, +1, +1\} \rightarrow \{+1, -1, +1\}$

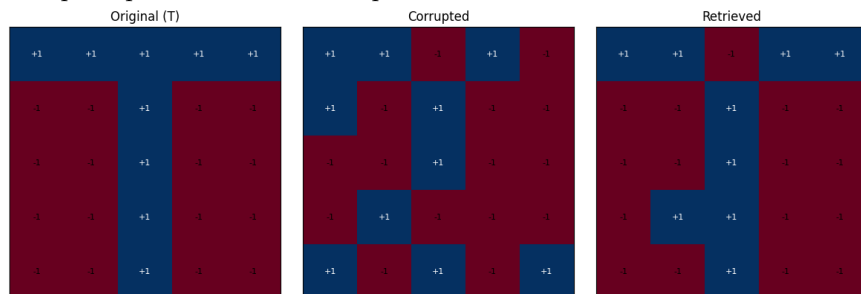
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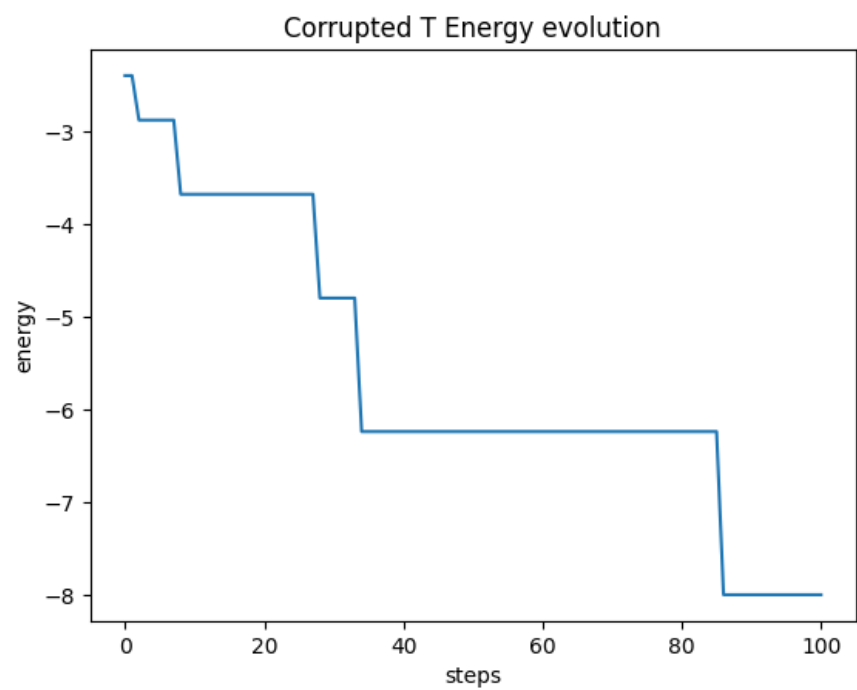
It's quite simple with just three neurons:



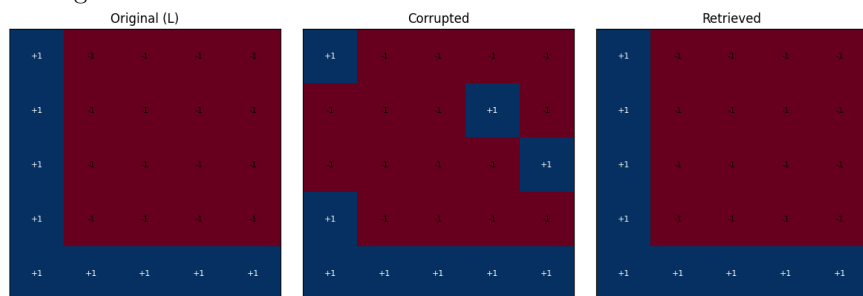
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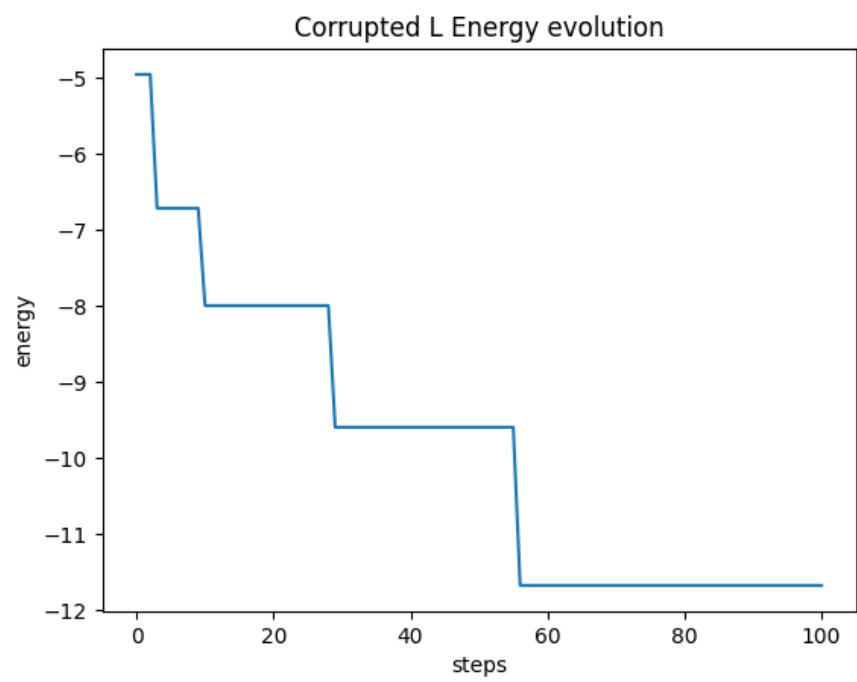
Let's step it up to memorize letter patterns.





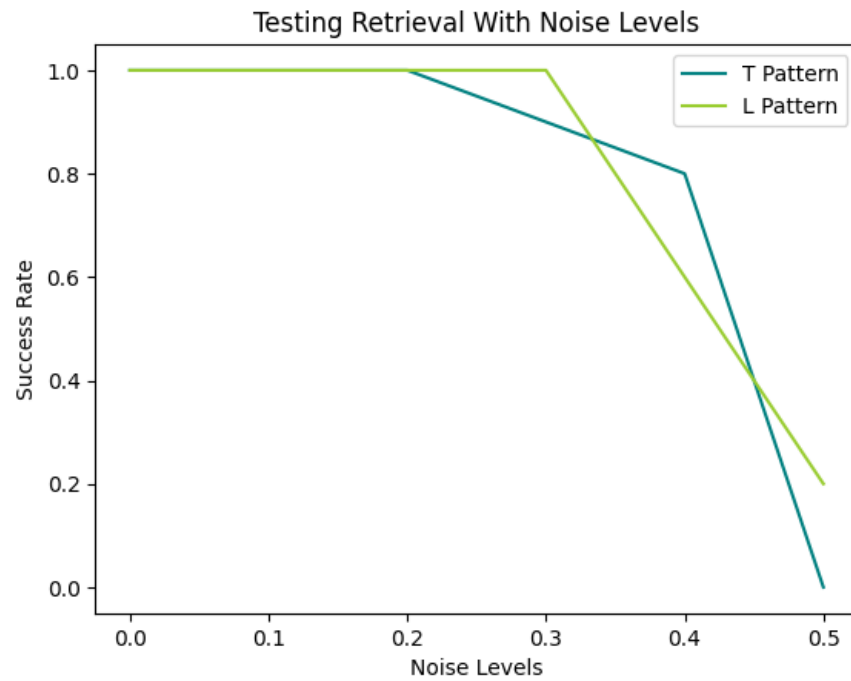
Testing another one.





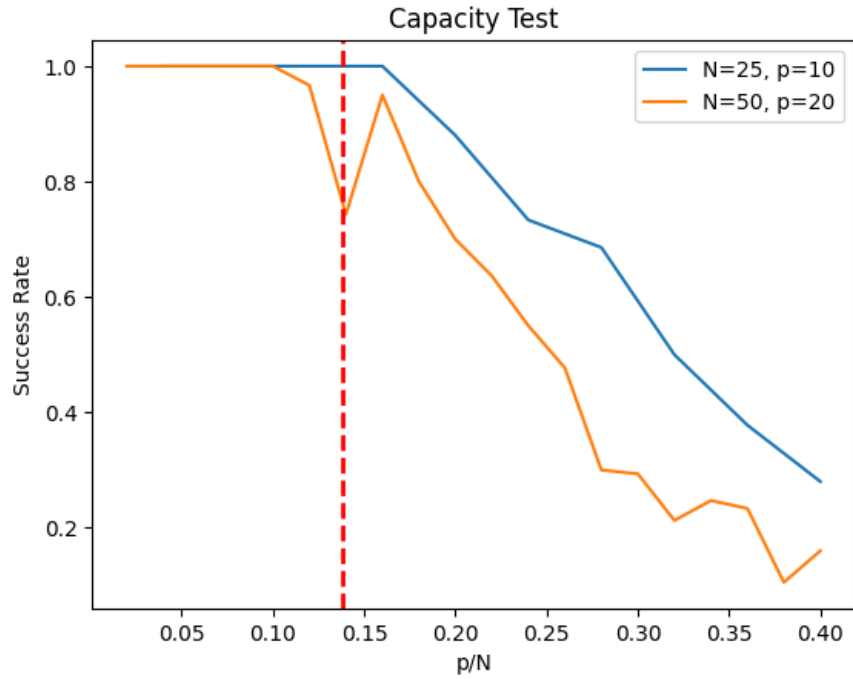
2.3: Testing More Properties

1



At 20% noise for the T pattern and 30% for L, the success rates start to drop off and retrieval fails.

2

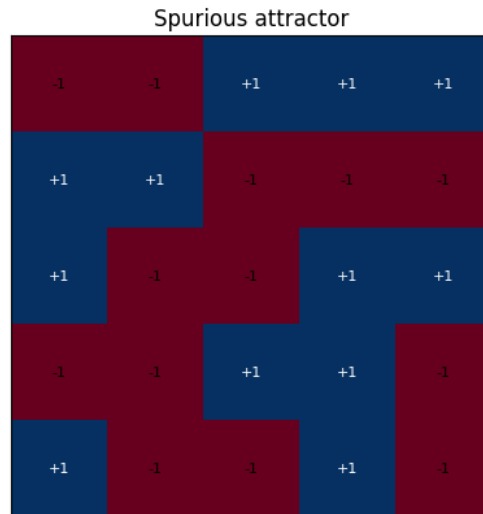


We plot p/N vs. Success Rate to see how the retrieval evolves at different ratios of number of patterns and network size. Both plots start to steadily drop off at $p/N \approx 0.16$ which is slightly higher than the theoretical limit (but differs with new random samples).

3

Num of Spurious Patterns: 9
 Num of Stored Patterns: 5

Sample spurious pattern:



With spurious attractors, we can't ensure every stable state in the network is truly a memory that we've added before, hurting reliability. This phenomenon occurs since we have more patterns to memorize than the capacity of the neurons, causing local minima to form in areas of interference. Depending on the initial locations of input patterns, Hopfield networks could permanently place the values into a spurious attractor which is a massive drawback to the deterministic process.

3.1: Limitations

1

Sparse connectivity in the brain means that the Hebbian learning rule we implemented to seed memories into the network isn't entirely plausible as the total view of each neuron isn't all the other neurons. However, even with limited connectivity, such a network could still have compositional/modular structure where groups of neurons hold a shared representation of activations. This could enable better expressability with stronger inter-neuron connections.

There are many interesting theories to explain how the brain could filter spurious patterns so well. This is clearly a big issue as the brain receives so many inputs from various contexts and conditions, yet we're still able to abstract away the details and reach a useful result.

- **Sparse coding:** If we have only a few neurons activate for each memory (instead of embedding the memory into every neuron's weight as done with Hopfield), we can minimize overlap between distinct memories. Funnily enough, artificial brains might be doing the same thing.
- **Pattern separator:** The primary issue with encoding multiple memories

is that they aren't orthogonal to each other, meaning new spurious activity is some form of linear combinations of the real patterns. There is a part in the brain called the dentate gyrus which can actually orthogonalize input patterns through its unique structure, creating high-dimensional representations with minimal overlap. What's more fascinating is the theory that neurogenesis might play a part: new, fresh neurons could encode new memories without interference of old patterns.

- **Sleep-dependent attractor basins:** During specific phases of sleep, a replay of stored memories could potentially prune the energy landscape by removing spurious patterns. This effect is possibly due to true memories synaptically becoming stronger due to replays during sleep, but spurious fake memories aren't shown and their basins become more shallow.

2

In a Hopfield network, storing multiple memories is computing the outer product of the desired states and averaging the weight matrices—in biological systems, this storage doesn't happen concurrently during training as the brain can create memories on the fly. This leads to the issue of catastrophic interference where earlier memories degrade as new ones are stacked on top. Hopfield assumes it's possible to globally sum over the patterns when setting the weights, but this is biologically implausible since each neuron can't sum over all other neurons with equal weight.

Biological neurons can't directly be used to minimize Hopfield's energy equation. Synaptic connections aren't symmetric, meaning that the weight matrix is a lot more complicated than the outer product of certain activations. Also, there isn't clearly a single state of the neuron at any point—instead it's an integration over timesteps of input history.

3.2: Beyond Hopfield

1

Restricted Boltzmann Machines (RBMs) are far more powerful than Hopfield networks, but they both fundamentally follow the same energy function:

$$E = - \sum_{i < j} W_{ij} s_i s_j - \sum_i \theta_i s_i$$

The latter seeks to store and retrieve memories while the former is building a probability distribution based on the inputs it was given. RBMs are a form of generative modeling, trying to model:

$$P(v, h) = \frac{1}{Z} e^{-E(v, h)}$$

Where v , h are states in the model and E is the energy function we defined earlier. RBMs are stochastic in nature, meaning a neuron's update isn't purely based on its weighted inputs but a probability that depends on the resulting energy difference and the temperature T (low $T \approx$ Hopfield and high T is almost random). Hence, this type of network assigns probability to all states depending on their energy value which differs from the discrete memories that HNs memorize. RBM's architecture is a two-layer, non-recurrent network. We lose the all-to-all connection but every hidden node is connected to every visible node, allowing for representational power (e.g. a hidden node might activate when specific pixels co-occur in the input). The fundamental new breakthrough that put "Restricted" in the name is only allowing connections between visible and hidden nodes and not within layers. Similar to the shortcut we took to approximate the stable state of the network by taking the outer product, here, each hidden unit is independent of one another which allows convergence with only one timestep.

2

Attractor networks with structured connectivity: The all-to-all connection regime that Hopfield networks impose is biologically implausible. Neural circuits are sparse, local, and hierarchical in certain locations and there are functional properties within groups and layers. The primary visual cortex has been interpreted to have neurons arranged on cortical sheets based on similar edge orientations. Instead of generic Hebbian connections from before, this region has neurons with shared orientation preferences excite each other and inhibit neurons with orthogonal preferences. This example shows how connectivity encodes prior knowledge about the structure of the representation space. Having modular structures give the ability to generalize continuously instead of being constrained by memorized discrete patterns. In fact, cortical networks can have multiple hierarchies of attractors with lower ones over local features and higher ones over compositions that lower attractors won't stabilize towards. Higher attractors can direct the lower attractors toward their end states.

4

1

We start with the notion of attractors and dynamical systems as a perspective to see how collective behavior is organized and executed. We focused on three types of continuous attractors:

1. Fixed points - The simplest form where the system moves monotonically through an energy landscape. We may have certain energy functions with multiple basins of attractions, thus some threshold determines where an initial condition will go.

2. Limit cycles - Instead of consistent change through a gradient, some systems will have oscillations where stable points can be determined by computing the Jacobian and finding the signs of eigenvalues.
3. Chaotic attractors - These are deterministic systems that change immensely with minor perturbations. The concept of criticality is vital when looking at certain behavior in groups of neurons (e.g. neural avalanches).

We shift our focus to discrete attractors like the Ising model where neighboring spins determine the overall energy and change necessary for the network. Again, when setting T to a specific number (2.27), we get interesting behavior in the network which ties back to the properties of chaotic attractors. The Ising model explicitly inspires Hopfield Networks which uses the concept of energy minima to store patterns and retrieve them from noisy cues. All these physics-based approaches fill in an aspect of the search towards neural computation, but still are limited in capability. One central example is that neurons aren't completely discrete magnets—somehow the brain operates with always-changing inputs. Also, Hopfield Networks lacked randomness which severely inhibited it from escaping local minima and evolve in the direction that may not most decrease total energy, an issue solved by RBMs and models thereafter.

2

This homework took around 8 hours. It was definitely the most intriguing one yet and I deeply enjoyed the programming and graphing. I much better understand where Hopfield was coming from and how his results laid out paths for further advancements. I was very unaware that attractors and dynamical systems from physics could explain so much about the brain's mechanics (especially the chaotic attractor), but there's still so much room to explore in terms of how different levels of attractors develop and how they couple with each other.

Code